

'Athlete's foot' (Tinea pedis) is caused by a number of different fungi, including species of Trichophyton, Epidermophyton, and Microsporum. It is a disease in which infection occurs generally between the toes.

4. Which of the following statements are correct?
Select the correct answer from the codes given below?

1. Femur is the longest bone in the human body.
2. Cholera is a disease caused by bacteria.
3. Athlete's foot is a disease caused by virus.

Code :

- (a) 1, 2 and 3 (b) 1 and 3
(c) 1 and 2 (d) 2 and 3

U.P.P.C.S. (Pre) 2008

Ans. (c)

The femur is the longest bone in the human body. Vibrio cholerae is the causing bacteria for cholera. The athlete's foot is an infectious disease caused by a number of different fungi, including species of Trichophyton, Epidermophyton, and Microsporum.

5. Which disease is caused by the fungi?

- (a) Polio (b) Dermatitis
(c) Cholera (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (b)

Seborrheic dermatitis disease is caused by the fungi. It is an inflammatory skin disorder affecting the scalp, face and torso. Polio is caused by a virus while cholera is a bacterial disease.

6. Which one of the following diseases is caused by Fungi?

- (a) Allergy (b) Colour blindness
(c) AIDS (d) Baldness

U.P. P.C.S. (Mains) 2016

Ans. (d)

Baldness is a fungal disease. AIDS is a fatal disease which affects the human immune system by HIV infection. Colour blindness is a genetic disorder.

7. Which of the following human organ is affected by the consumption of aflatoxin food adulterant ?

- (a) Heart (b) Lungs
(c) Kidney (d) Liver

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (d)

The liver is major digestive gland which is affected by aflatoxins. Some fungal species of Aspergillus and Penicillium produce aflatoxins that make food toxic.

IV. Other Parasitic Diseases

Notes

- Besides virus, bacteria and fungus some members of phylum-protista, platyhelminthes and nematoda are responsible for different diseases.
- Some parasites like Toxoplasma gondii and Plasmodium spp. can cause disease directly, but other organisms can cause disease by the toxins that they produce.
- The most important parasitic diseases are as follows—

Malaria :

- Malaria is a life-threatening disease. It is typically transmitted through the bite of female **Anopheles** mosquito. Infected mosquitoes carry the **Plasmodium** parasite. When this mosquito bites, the parasite is released into the bloodstream. The primary host of Plasmodium is a man and the secondary host is female Anopheles.
- Once the parasites are inside the body, they travel to the liver, where they mature. After several days, the mature parasite enters the bloodstream and begin to infect **red blood corpuscles** (RBCs).
- Within 48 to 72 hours, the parasites inside the red blood corpuscles multiply, causing the infected cells to burst open.
- The parasites continue to infect red blood cells, resulting in symptoms that occur in cycles that last two to three days at a time.
- Malaria is typically found in tropical and subtropical climates where the parasites can live.
- There are four kinds of malaria parasites that can infect humans : **Plasmodium vivax**, **P. ovale**, **P. malariae** and **P. falciparum** (unicelled protozoan).
- **P. falciparum** causes a more severe form of the disease (**malignant malaria**) and those who contract this form of malaria have a higher risk of death.
- An infected mother can also pass the disease to her baby at birth. This is known as **congenital malaria**.
- Malaria is transmitted by blood, so it can also be transmitted through –
 - (1) An organ transplant
 - (2) A transfusion
 - (3) Use of shared needles or syringes
- The symptoms of malaria typically develop within 10 days to 4 weeks following the infection. In some cases,

symptoms may not develop for several months. Some malarial parasites can enter the body but will be dormant for long periods of time.

- Common symptoms of malaria include shaking chills that can range from moderate to severe, high fever, profuse sweating, headache, nausea, vomiting etc.
- Malaria can cause a number of life-threatening complications. The following may occur :
 - (1) Swelling of the blood vessels of the brain or cerebral malaria.
 - (2) An accumulation of fluid in the lungs that causes breathing problems or **pulmonary oedema**.
 - (3) Organ failure of the kidney, liver or spleen.
 - (4) Anaemia due to the destruction of RBCs.
 - (5) Low blood sugar.
- **Ronald Ross** was a British Medical doctor who received the Nobel Prize for **Physiology** or **Medicine** in 1902 for his work on the transmission of malaria. His discovery of the malarial parasite in the gastro-intestinal tract of mosquito in 1897 proved that malaria was transmitted by mosquitoes, and laid the foundation for the method of combating the disease.
- Malaria can be prevented by avoiding mosquito bites by using mosquito repellants (Pyrethrum) and nets.
- Quinine is a medication used to treat malaria. It was first isolated in 1820 from the bark of *cinchona tree*.
- Some antimalarial drugs as chloroquine, amodiaquin, atovaquone, proguanil, sulphonamides, mefloquine, primaquine are also used to cure malaria.
- Use of antimalarial drugs to prevent the development of malaria is known as chemoprophylaxis. These drugs should be used regularly in malaria affected regions.
- To eradicate the Anopheles mosquito, pesticides such as DDT should be used.
- Mosquito larvae eating fish **Gambusia affinis** is used to control the population of mosquito.
- The ducks are also helpful in eradicating the larvae of mosquito.
- **World Malaria Day** is an international observance commemorated every year on 25 April and recognizes global efforts to control malaria.

Kala-azar (Visceral leishmaniasis) :

- Kala-azar is a disease caused by a protozoan parasite of the genus **Leishmania**. It is a chronic and potentially fatal disease chiefly of tropical areas.
- The parasite primarily infects the reticuloendothelial system and may be found in abundance in bone marrow, spleen and liver.

- The signs and symptoms of kala-azar is light coloured persons show greyish discolouration of the skin of hands, feet, abdomen and face which gives the Indian name **Kala-azar** meaning **Black fever**. Some symptoms may include fever, loss of weight and inflammation of liver and spleen.
- The parasite lives and multiplies inside the female **sandfly (Phlebotomus argentipes)**. This insect is most active in humid environments during the warmer months and at night from dusk to dawn. Domestic animals, such as dogs, can serve as reservoirs for the parasite. Transmission may occur from animal to sandfly to human.
- Humans can also transmit the parasite between each other through a **blood transfusion** or shared needles.
- This disease is the second-largest parasitic killer in the world (after malaria).

Sleeping Sickness :

- An infectious disease, especially common in tropical Africa, caused by either of two trypanosomes (**Trypanosoma gambiense** or **Trypanosoma rhodesiense**) that are transmitted by the bite of the tsetse fly (genus Glossina). This disease is also known as African trypanosomiasis.
- It is characterized by fever, drowsiness, and coma, usually ending in prolonged coma and death.
- Sleeping sickness is characterized by two stages of illness. In the first stage, infected persons typically experience fever, headache, muscle and joint pain and inflammation of the **lymph nodes**. The second stage, which develops within several weeks (T- rhodesiense) or within one or two years (T. gambiense), is marked by involvement of brain and spinal cord, accompanied by personality changes, sleep disturbances and profound lethargy (a lack of energy and enthusiasm), frequently ending in death when left untreated.
- Trypanosoma is a member of phylum Protozoa.

Amoebiasis (Amoebic Dysentery) :

- Amoebiasis is caused by an anaerobic parasite amoeba- **Entamoeba histolytica** (Phylum-Protozoa).
- Amoebiasis can be present with no, mild or severe symptoms. Symptoms may include abdominal pain, diarrhea, or bloody diarrhea.

Schistosomiasis :

- Schistosomiasis is also known as **Snail fever** and **bilharzia**.
- It is caused by parasitic flatworms called **schistosomes**.
- The urinary tract or intestines may be infected.
- Symptoms include abdominal pain, diarrhea, bloody stool or blood in urine.

- In developing countries due to unsafe potable water and lack of sanitation, the **malaria**, **dysentery** and **schistosomiasis** have taken the form of the **epidemic** (a widespread occurrence of an infectious disease in a community at a particular time).

Fascioliasis :

- Fascioliasis (Liver rot) is a parasitic worm infection caused by liver flukes viz. **Fasciola hepatica** and **Fasciola gigantica**.
- It affects humans, but its main host is ruminants such as cattle and sheep.
- People usually become infected by eating raw watercress or other watergrown plants contaminated with immature parasite larvae.
- The young worms move through the intestinal wall, the abdominal cavity and the liver tissue, into the bile ducts.
- The symptoms of fascioliasis in humans are fever, sweating, loss of weight, abdominal pain.

Neurocysticercosis :

- **Neurocysticercosis** is the result of accidental ingestion of eggs of **Taenia solium** (i.e. pork tapeworm), usually due to contamination of food by people with taeniasis.
- It is the parasitic disease of the central nervous system and is the main cause of acquired epilepsy.
- It is a leading cause of seizures and epilepsy in the developing world and most common parasitic disease of the central nervous system.
- Tennis champion, **Leander Paes** was suffering from a parasitic infection of **Taenia solium**.
- **T. solium** infection can be transmitted by consumption of raw and/or undercooked pork product or contaminated water with its eggs.

It is noteworthy that **Fasciola**, **Schistosoma** and **Taenia** are the members of Phylum **Platyhelminthes**.

In human body approximately 50 species of phylum **Nematoda** (roundworms) are found as parasitic worm causing different diseases.

Filariasis :

- Filariasis is a parasitic disease caused by an infection with roundworm named **Wuchereria bancrofti**.
- Filariasis is also known as **elephantiasis**. In elephantiasis the lymph glands become swollen.
- **W. bancrofti** can affect the legs, arms, vulva, breasts and scrotum (causing **hydrocele** formation).
- The female **Culex** mosquito acts as a vector in the transmission of this disease.

Ascariasis :

- Ascariasis is a disease (infection of small intestine) caused by the parasitic roundworms **Ascaris lumbricoides**.
- A person can become infected with **ascariasis** after accidentally ingesting the eggs of **A. lumbricoides** roundworm.
- The eggs can be found in soil contaminated by human feces or uncooked food contaminated by soil that contains roundworm eggs.
- Children often become infected when they put their hands in their mouths after playing in contaminated soil.
- The symptoms of ascariasis are-
(i) coughing or gagging
(ii) wheezing or shortness of breath
(iii) blood in mucus
(iv) chest discomfort
(v) abdominal swelling and pain, diarrhea
- Roundworm can cause nausea, vomiting, loss of appetite, and growth impairment in children due to **malabsorption**.
- **Anthelmintic** medications (drugs that rid the body of parasitic worms) such as **chenopodium oil**, **albendazole** and **mebendazole** are the drugs of choice for the treatment of **Ascariasis**.

Question Bank

1. The organ affected by malaria is –

- (a) Heart (b) Lungs
(c) Kidney (d) Spleen

M.P.P.C.S. (Pre) 1999

Ans. (d)

Malaria is transmitted most commonly by Plasmodium parasite whose carrier is female Anopheles mosquito. The spleen, which often becomes enlarged in malaria, is most affected part of the body. Quinine is the age-old traditional drug for treatment of malaria. Other effective drugs are camaguin, daprim, resochin, etc.

2. Carrier of parasite Plasmodium is :

- (a) Tick (b) Mosquito
(c) Fly (d) Lice

R.A.S./R.T.S.(Pre) 2010

Ans. (b)

See the explanation of above question.

3. Which of the following is a carrier of parasite Plasmodium?

- (a) Mosquito (b) Fly
(c) Lice (d) All of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

See the explanation of above question.

4. Besides spreading Malaria, Anopheles mosquito's are also vectors of :

- (a) Dengue and Filariasis
- (b) Dengue and Encephalitis
- (c) Encephalitis and Filariasis
- (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

While Anopheles mosquitoes are well-known as vectors for malaria, they can also be vectors for certain types of encephalitis. Japanese Encephalitis Virus (JEV) is a significant cause of encephalitis in Asia, and while Culex mosquitoes are the primary vectors, Anopheles mosquitoes, such as Anopheles subpictus, have been identified as vectors in specific regions where they are more abundant. Anopheles mosquitoes can also carry other encephalitis viruses, such as St. Louis Encephalitis Virus (SLEV) and Eastern Equine Encephalitis Virus (EEE). Anopheles mosquitoes are also vectors of filariasis in some regions, particularly in rural areas of Asia and Africa. In fact, the World Health Organization (WHO) notes that lymphatic filariasis is transmitted by various mosquito species, including Anopheles. Other important vectors for filariasis include Culex and Aedes mosquitoes. Hence, option (c) is the correct answer.

5. Malignant malaria is caused by :

- (a) Plasmodium vivax
- (b) Plasmodium ovale
- (c) Plasmodium malariae
- (d) Plasmodium falciparum

R.A.S./ R.T.S. (Pre) 2021

Ans. (d)

Malignant malaria is caused by Plasmodium falciparum. It is among the most severe human infectious diseases and those who contract this form of malaria have a higher risk of death.

6. Species of Plasmodium which causes Malignant Malaria :

- (a) Vivax
- (b) Falciparum
- (c) Ovale
- (d) Malarial

Rajasthan P.C.S. (Pre) 2023

Ans. (b)

See the explanation of above question.

7. Which one of the following is NOT common between Malaria and Dengue?

- (a) Fever
- (b) Mosquito bite
- (c) Human species
- (d) Mosquito species

U.P.P.C.S. (Pre) 1998

Ans. (d)

Malaria is transmitted by the bite of female Anopheles mosquito while dengue is from female Aedes aegypti mosquito. Except this, Elephantiasis and yellow fever is also spread by mosquito.

8. Which of the following diseases is caused by protozoa?

- (a) Cholera
- (b) Diphtheria
- (c) Pneumonia
- (d) Malaria

M.P.P.C.S. (Pre) 2017

Ans. (d)

Malaria is caused by a protozoa parasite called plasmodium and its carrier is female anopheles mosquito, which is secondary host.

9. Which disease among these is spread by a protozoa ?

- (a) Malaria
- (b) Cholera
- (c) Jaundice
- (d) None of above

U.P.P.C.S. (Mains) 2008

Ans. (a)

See the explanation of above question.

10. Which statement is not true regarding malaria?

- (a) It is a disease caused by insect parasite
- (b) It is spread by the mosquitoes
- (c) It spreads mostly in swampy region
- (d) For its treatment chloroquine is used

38th B.P.S.C. (Pre) 1992

Ans. (a)

Malaria disease is caused by plasmodium parasite which is a protozoa. Thus option (a) is not true. Other three options are correct.

11. Which of the following parasites is responsible for 65% of the cases of malaria in India ?

- (a) P. malariae
- (b) P. vivax
- (c) P. falciparum
- (d) P. ovale

U.P.P.C.S. (Mains) 2008

Ans. (b)

Plasmodium vivax is a protozoan parasite and a human pathogen, the most frequent and widely distributed cause of recurring malaria. P. vivax is one of the five species of malaria parasites and it is responsible for about 65% cases of malaria in India. Actually Plasmodium vivax accounts for 65% of malaria cases in Asia and South America.

12. Who was the scientist who discovered that malaria is caused by plasmodium parasite?

- (a) J.G. Mendel
- (b) Haeckel
- (c) Sir Ronald Ross
- (d) Darwin

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (c)

Sir Ronald Ross, a doctor in British Indian Army first observed oocysts of plasmodium in the female Anopheles mosquito. He won the Nobel prize in 1902 for his work on malaria.

13. What is R21/Matrix-M™?

- (a) Vaccine against Malaria developed by Serum Institute of India
- (b) Vaccine against Malaria developed by Bharat Biotech International
- (c) Vaccine against AIDS developed by Serum Institute of India
- (d) Vaccine against AIDS developed by Bharat Biotech International

U.P.P.C.S. (Pre) 2024

Ans. (a)

The R21/Matrix-M™ vaccine, developed through collaboration between the Jenner Institute at Oxford University and the Serum Institute of India leveraging by Novavax's saponin-based adjuvant technology, is the second malaria vaccine to be authorized for use in children in malaria-endemic regions. R21/Matrix-M™ vaccine received WHO recommendation for use in children in October, 2023. RTS, S/AS01 (also known as Mosquirix), a recombinant protein-based malaria vaccine, was the first malaria vaccine approved by the WHO.

14. The diseases caused by nematodes :

- (a) Filaria
- (b) Fluorosis
- (c) Encephalitis
- (d) Leprosy

U.P.P.C.S. (Pre) 1999

Ans. (a)

Filariasis is a parasitic infection caused by thread-like nematodes (filariae) that belong to the roundworm superfamily Filarioidea. These infestations are common in tropical countries such as sub-Saharan Africa, southern Asia, the western Pacific Islands, Brazil and Guyana. The disease spreads from person to person by female Culex mosquito. Edema is one of the most common symptoms of filariasis and can lead to elephantiasis, which is characterized by thickened skin and massive swelling.

15. Select the incorrect set of “the typical Vector – Disease caused – Type of Pathogen”.

- (a) Aedes – Chikungunya – Bacteria
- (b) Culex – Lymphatic Filariasis – Parasite
- (c) Anopheles – Malaria – Parasite
- (d) Aedes – Dengue – Virus

Rajasthan P.C.S. (Pre) 2024

Ans. (a)

Among the given options, set of option (a) is incorrect, as chikungunya is a viral disease transmitted to humans through by Aedes mosquitoes infected with the chikungunya virus (CHIKV). CHIKV is a single-stranded RNA virus in the alphavirus genus of the family Togaviridae. Sets of other three options are correct.

16. Match List-I with List-II and select the correct answer using the codes given below :

List-I (Disease)	List-II (Organism)
A. Malaria	1. Fungi
B. Poliomyelitis	2. Bacteria
C. Tuberculosis	3. Virus
D. Ringworm	4. Protozoan

Code :

	A	B	C	D
(a)	4	3	2	1
(b)	4	3	1	2
(c)	3	4	1	2
(d)	3	4	2	1

I.A.S. (Pre) 1998

Ans. (a)

The correctly matched lists are as follows :

Malaria	–	Protozoan
Poliomyelitis	–	Virus
Tuberculosis	–	Bacteria
Ringworm	–	Fungi

17. Match List-I with List-II and select the correct answer from the codes given below the lists :

List-I	List-II
A. Plague	1. Protozoa
B. AIDS	2. Fungus
C. Baldness	3. Virus
D. Malaria	4. Bacteria

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	4	1
(c)	3	4	1	2
(d)	4	3	2	1

U.P.P.C.S. (Pre) 2012

Ans. (d)

The correctly matched lists are as follows :

Plague	–	Bacteria
AIDS	–	Virus
Baldness	–	Fungus
Malaria	–	Protozoa

18. Match List-I with List-II and select the correct answer using code given below :

List-I (Diseases)	List-II (Factors)
A. Peptic ulcers	1. Virus
B. Dengue	2. Protozoa
C. Elephant Foot	3. Worm
D. Malaria	4. Bacteria

Code :

	A	B	C	D
(a)	4	2	3	1
(b)	3	4	2	1
(c)	4	1	3	2
(d)	2	3	4	1

U.P. P.C.S. (Pre) 2023

Ans. (c)

The correctly matched lists are as follows :

Diseases	Factors
Peptic ulcers	— Bacteria
Dengue	— Virus
Elephant Foot	— Worm
Malaria	— Protozoa

19. Sleeping sickness happens due to :

- Deficiency of Vitamin-A
- Deficiency of calcium in body
- Rising of blood pressure
- Unicellular animal named Trypanosoma

U.P.P.C.S. (Mains) 2007

Uttarakhand U.P.A/L.D.A. (Pre) 2003

Ans. (d)

Sleeping sickness, also called 'human African trypanosomiasis' is a widespread tropical disease that can be fatal if not treated. Sleeping sickness is caused by two different parasites, *Trypanosoma brucei rhodesiense* and *Trypanosoma brucei gambiense*. It is spread by the bite of an infected tse-tse fly (genus *Glossina*), a species native to the African continent. The tse-tse fly bite erupts into a red sore and within a few weeks, the person can experience fever, swollen lymph glands, aching muscles and joints, headaches and irritability.

20. Which of the following is NOT mosquito borne disease?

- Dengue fever
- Filariasis
- Sleeping sickness
- Malaria

U.P. P.C.S. (Pre) 2021

Ans. (c)

Mosquito-borne diseases are those spread by the bite of an infected mosquito. Diseases that are spread to people by mosquitoes include Zika virus, West Nile virus, Chikungunya virus, dengue, and malaria.

Sleeping sickness, or human African trypanosomiasis, is a life-threatening disease caused by related parasite strains, *Trypanosoma brucei gambiense* and *Trypanosoma brucei rhodesiense*, transmitted by the tse-tse fly.

21. Due to unsafe drinking water and lack of sanitation, three main communicable diseases in developing countries are –

- Diarrhoea, cancer, gout
- Malaria, Acute diarrhoea, Schistosomiasis
- Orchociasis, white blindness, arthritis
- Rheumatism, Malaria, AIDA

I.A.S. (Pre) 1993

Ans. (b)

Malaria, Acute Diarrhoea and Schistosomiasis are the three main communicable diseases in developing countries caused by unsafe water and lack of sanitation.

22. Consider the following statements :

- Taeniasis is an intestinal infection caused by three species of tapeworm – *Taenia solium*, *Taenia saginata* and *Taenia asiatica*.
- When cysts develop in the brain, the condition is referred to as neuro-cysticercosis (NCC).

Which of the above statements is/are correct?

- Only 1
- Only 2
- Both 1 and 2
- None of the above

69th B.P.S.C. (Pre) 2023

Ans. (c)

The term taeniasis refers to intestinal infection with tapeworms. Three parasite species cause taeniasis in humans. *Taenia solium* (pork tapeworm), *Taenia saginata* (beef tapeworm) and *Taenia asiatica* (Asian tapeworm). Only *T. solium* causes major health problems. *T. solium* taeniasis is acquired by humans through the ingestion of the parasite's larval cysts (cysticerci) in undercooked and infected pork. Humans can also become infected with *T. solium* eggs due to poor hygiene (via the fecal-oral route) or ingesting contaminated food or water. Ingested *T. solium* eggs develop to larvae (called cysticerci) in various organs of the human body. When they enter the central nervous system (i.e. in brain and spinal cord), they can cause neurological symptoms (neurocysticercosis), including epileptic seizures. Hence, both statements are correct.

23. Leander Paes was forced off the Court of Tennis in 2003 after a problem was detected in his brain. What was the problem ?

- A tumour
- A haemorrhage
- Some spinal fluid
- A cyst

U.P.P.C.S. (Mains) 2005

Ans. (d)

Leander Paes was forced off the Court of Tennis in 2003 after a cyst was detected in his brain. A cyst is a closed sac, having a distinct membrane and division compared to the nearby tissue. Hence, it is a cluster of cells that have grouped together to form a sac. One rather common infection in India causing such a brain lesion is cysticercus granuloma caused by the larval form of a tapeworm.